

and local folk medicine traditionally considered the plant a treatment for toothache. Coca leaf extract is still used as a flavoring for cola drinks—but cocaine has long been banned from the formulas.

Medicinal Actions & Uses In Bolivia and Peru, coca leaves play an important part in the culture and herbal medicine of the indigenous Aymara and Quechua peoples. High altitudes, cold, and an impoverished diet place great physical demands on the population. Coca leaves, chewed with lime or ashes, release small amounts of the active constituents, which act as a tonic and help block the effects of cold, exhaustion, and poor nutrition. Coca leaves are also used in South American herbal medicine to treat nausea, vomiting, and asthma, and have been used to speed convalescence. Cocaine extracted from coca leaves is used legally in conventional medicine as a local anesthetic. It is also taken illegally as a narcotic, stimulant drug. As an isolated chemical, cocaine is extremely addictive.

⚠ Cautions Take only under professional supervision. Coca is subject to legal restrictions in most countries.



Coca is cultivated in the Andes and chewed as a tonic to help counter the effects of cold.

Eschscholzia californica (Papaveraceae) California Poppy

Description Annual or perennial growing to 2 ft (60 cm). California poppy has finely cut leaves and bright orange, yellow, pink, or red flowers.

Habitat & Cultivation California poppy is native to western North America. Widely cultivated as a garden plant, it prefers sandy soils.

Parts Used Aerial parts.

Constituents California poppy contains isoquinoline alkaloids (including protopine, cryptopine, and chelidonine) and flavone glycosides.



California poppy contains a latex that has sedative, pain-killing, and antispasmodic properties. It is a gentle remedy, suitable for children.

History & Folklore Native American peoples were known to use the sap of California poppy for its pain-killing properties, particularly for toothache. The leaves were also eaten as a vegetable. Early settlers used California poppy for sleep problems, especially in children, and for whooping cough. It is California's state flower.

Medicinal Actions & Uses Though the California poppy is a close relation of the opium poppy (*Papaver somniferum*, p. 244), it has a significantly different effect on the central nervous system. California poppy is not a narcotic. In fact, rather than disorienting the user, it actually tends to normalize psychological function. California poppy's gently antispasmodic, sedative, and analgesic effects make it a valuable herbal medicine for treating physical and psychological problems in children. The herb may prove beneficial in attempts to overcome nervous tension and anxiety, bedwetting, and difficulty in sleeping.

Research A French study confirmed the traditional usage of California poppy by showing that extracts were sedative, reduced anxiety levels, and were non-toxic.

Eucalyptus smithii (Myrtaceae) Gully Gum

Description Aromatic evergreen tree growing to 165 ft (50 m).

Habitat & Cultivation Native to Australia, eucalyptus now grows in temperate and subtropical zones across the globe. It prefers moist soil, and so is found in swamps, gullies, and at the foot of slopes.

Part Used Essential oil.

Constituents The volatile oil contains about 70% eucalyptol (1, 8-cineole), as well as pinene, limonene, alpha-terpineol, and linalool. While it is similar to the oils of related species, this oil appears to be better tolerated by the skin.

Medicinal Actions & Uses *Eucalyptus smithii* oil is an antiseptic and decongestant, and is used for inhalations as well as aromatherapy massages. See eucalyptus (*Eucalyptus globulus*, p. 96) for further details.

Cautions Though less toxic than the oils of other eucalyptus species, *E. smithii* essential oil should be used with care. Follow the label instructions, or take on professional advice.

Eucommia ulmoides (Eucommiaceae) Eucommia Bark, Du Zhong (Chinese)

Description Deciduous tree growing to 65 ft (20 m). Has elliptical leaves, with male flowers in loose clusters and solitary female flowers in the leaf axils.

Habitat & Cultivation Eucommia bark grows in temperate zones in China. It is cultivated, but only in small amounts.

Part Used Bark.

Constituents Eucommia bark contains gutta-percha, alkaloids, flavonoids, iridoids and other glycosides, and phenolic compounds.

History & Folklore The herb is mentioned in the Chinese herbal, the *Divine Husbandman's Classic* (*Shen'ngong Bencaojing*), which was written in the 1st century CE.

Medicinal Actions & Uses Eucommia bark is considered an excellent tonic for the liver and kidneys. Eucommia bark is said to "tonify the yang," to improve the circulation, and also to prevent miscarriage in women who are weak or suffering from back pain.